

Sequences and series



1 A sequence is a list of terms, and a series is the sum of the terms.

2 6

3 196

4 a_5

5 a Finite b Infinite c Finite d Infinite

6 Finite

7 a Infinite b Finite

8 a 0, 3, 6, 9, 12 b $-2, -4, -8, -16, -32$
c 4, 8, 16, 32, 64 d $-6, 24, -72, 192, -480$
e 1, 0.7071, 0.5774, 0.5, 0.4472 f 2, 2.25, 2.3704, 2.4414, 2.4883

9 a $\frac{5}{4}, \frac{6}{5}, \frac{7}{6}, \frac{8}{7}$ b $\frac{2}{3}, 1, 1, \frac{19}{21}$ c $-1, -\frac{1}{4}, 0, \frac{1}{16}$ d 0, -1, 2, -3

10 a 4, 3, 4, 11 b 3, 5, 8, 12 c $-2, -2, 0, 8$ d 3, 5, 15, 75
e 3, 2, 0, -8



Sigma notation

11

a

$$\sum_{k=1}^{\infty} 4k$$

b

$$\sum_{k=1}^6 (-3)^k$$

c

$$\sum_{k=1}^5 \frac{1}{k^3}$$

d

$$\sum_{k=1}^7 \frac{1}{3k}$$

e

$$\sum_{k=1}^{11} \frac{2}{5+k}$$

f

$$\sum_{k=1}^{11} \left(-\frac{1}{2}\right)^{k-1}$$

12

a

40

b

$$\frac{47}{60}$$

c

$$\frac{19}{20}$$

d

15

e

14

f

1815

g

368

h

-16

i

$$\frac{63}{2}$$

j

38

13 20

14 -10

15 30

16 -8

17

a

-6

b

-8

18

a

-54

b

65

19



a

$$\sum_{i=1}^7 (i + i^3) = 2 + 10 + 30 + 68 + 130 + 222 + 350$$

$$\begin{aligned} \sum_{i=1}^7 i + \sum_{i=1}^7 i^3 &= (1 + 2 + 3 + 4 + 5 + 6 + 7) + (1 + 8 + 27 + 64 + 125 + 216 + 343) \\ &= 2 + 10 + 30 + 68 + 130 + 222 + 350 \end{aligned}$$

b Yes

c Yes

Applications

20 $m_1 = 2000, m_n = (0.76) m_{n-1}$

21

a \$1, \$3, \$7, \$15

b \$38, \$43, \$48, \$53

c Day 6